Tectonophysics, 98 (1983) 271–295
Elsevier Science Publishers B.V., Amsterdam – Printed in The Netherlands

A GEOLOGICAL AND PALEOMAGNETIC STUDY OF THE HYBLEAN VOLCANIC ROCKS, SICILY

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(Received November 23, 1982; revised version accepted February 22, 1983)

ABSTRACT

Grasso, M., Lentini, F., Nairn, A.E.M. and Vigliotti, L., 1983. A geological and paleomagnetic study of the Hyblean volcanic rocks, Sicily. *Tectonophysics*, 98: 271–295.

The geological and paleomagnetic study of the Hyblean volcanic rocks in southeastern Sicily permits the following conclusions:

(1) The Upper Cretaceous volcanics are considerably more extensive than earlier studies indicated, they stretch from north of Augusta to the vicinity of Siracusa. Their magnetization is identical to that of the better known outcrops around Capo Passero. Of the two new age dates reported here, one is younger than any so far reported.

(2) The Upper Tortonian flows are normally magnetized, suggesting a correlation of the Carlentini Formation with magnetic interval 7. The flows at the top of the Tellaro Formation are also normally magnetized and are tentatively correlated with the Carlentini Formation flows. Shorter reversed intervals within interval 7 may be represented by two less well-dated flows.

(3) The lowest Pliocene flows have an adequate stratigraphic control and their magnetization is consistent with the extrusion of the pillow lavas during the Mammoth event. The Militello flow sequence together with the Francofonte flows appear by indirect argument to lie within the oldest reversed interval, i.e. pre-Kaena, of the Matuyama epoch. The individual flows in which both normal and reversed directions have been found cannot be assigned positions in magneto-stratigraphic scale.

(4) The geomagnetic pole positions are consistent with that of the African plate during the Cretaceous, and a European plate location during both Pliocene and Miocene time. The pole positions are:

Upper Cretaceous	212.3°E, 69.3°N.
Upper Miocene:	200.9°E, 82.1°N.
Pliocene	339.3°E, 85.9°N.

INTRODUCTION

Nearly a decade has passed since the first paleomagnetic studies were made in southeastern Sicily by Schult (1973), Barberi et al. (1974), and Gregor et al. (1975). In those papers particular attention was given to the Cretaceous data, for these data

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were consistent with the location of this part of Sicily on the African plate. Less attention was given to the younger volcanics (Barberi et al., 1974, referring only to the polarity of the samples), which according to the K/Ar ages are mostly of Pliocene age. No paleomagnetic work appears to have been attempted on the Miocene volcanics, perhaps because the mode of occurrence of the rocks is not encouraging (Rittman, 1973; Cucozzo-Silvestre, 1961, 1963).

In the intervening decade considerable progress has been made in the detailed stratigraphy of the Hyblean Plateau, through the work of Grasso et al. (1979, 1982), Patacca et al. (1980), Pedley (1981), and Grasso and Lentini (1982). One consequence of this is that the stratigraphic location and distribution of the volcanic horizons is much better known, which suggested a profitable feed-back from renewed magnetic study.

The region is important since it is one of the few areas in southern Italy where a stable craton can be recognized and hence used as a reference point in tectonic

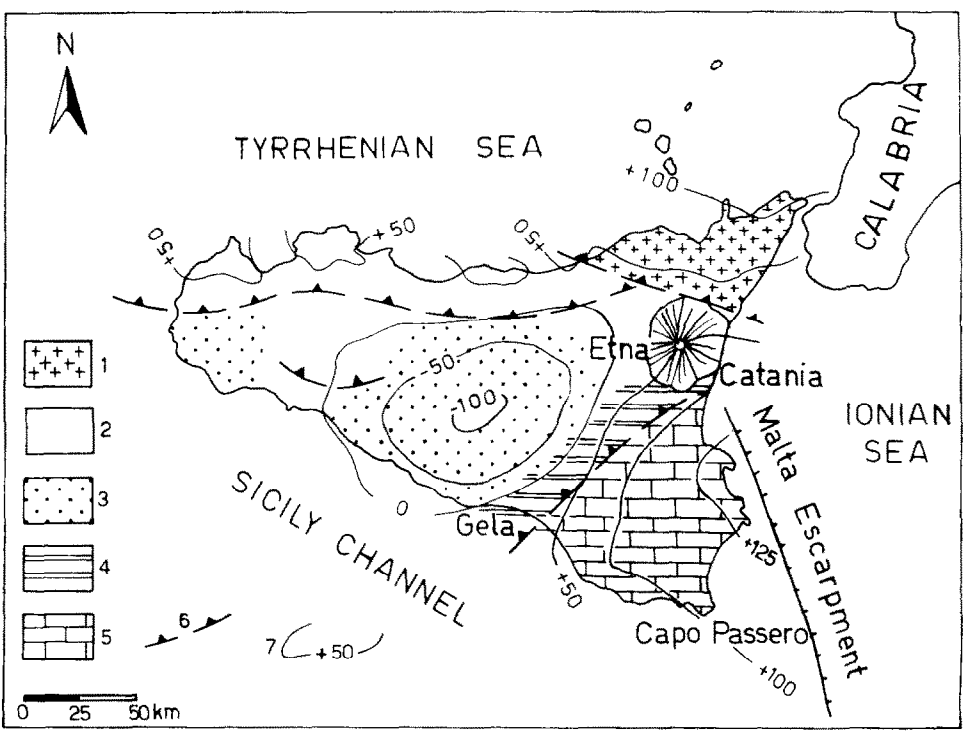


Fig. 1. The principal structural elements of Sicily. *1* = Peloritani–Calabrian internal thrust units, *2* = Maghrebian thrust sheet, *3* = Caltanissetta and Castelvetrano basins, *4* = Gela–Catania foredeep, *5* = Hyblean Plateau, *6* = fronts of the main thrust sheets, *7* = Bouguer anomaly contours (simplified after A.G.I.P., 1978). Note that the Hyblean Plateau is cut off to the east by the faults which form the modern Malta Escarpment, and to the north by the Gela–Catania foredeep of Plio-Pleistocene age.

studies, particularly in the interpretation of discordant paleomagnetic data (see, for example, Catalano et al., 1983).

For the purposes of this study of the seemingly limitless possibilities for sampling, a selection of over 170 sites was based primarily upom stratigraphically well-defined flows and in obtaining a well spaced network of sites (Fig. 1). A small number of sites whose stratigraphic situation is less well defined were collected in order to attempt radiometric dating valuable for stratigraphic purposes. Attention was also given to some of the sections described by Honnorez (1962), where it was possible to collect from flow sequences.

GENERAL GEOLOGY

The Hyblean (or Iblean) Plateau occupies the southeastern corner of Sicily. It is a segment of stable foreland separated from the zone of thrusts to the north by the Gela–Catania foredeep (Fig. 1).

Formerly more extensive, it is now cut off to the east by the faults of the Malta Escarpment which separate it from the modern Ionian abyssal plain.

Throughout most of the Mesozoic (Patacca et al., 1979) and Cenozoic (Grasso and Lentini, 1982) carbonate sedimentation predominated over the Hyblean Plateau. Information about all horizons below the Upper Cretaceous is available only from boreholes.

A recent interpretation of the subsurface data by Patacca et al. (1979) indicates that an extensional tectonic regime affected the Hyblean foreland in early Jurassic time, and was renewed both after middle Jurassic and in Senonian time. These authors could also distinguish distinctive facies in the two regions, the Siracusa and Ragusa belts, described by the early petroleum geologists (Colombo, 1969). These zones maintained their individuality late into Cenozoic time (Grasso and Lentini, 1982) and were important in the development of the Miocene paleogeographic pattern and of particular interest here because of the volcanic developments.

Volcanic activity seen in outcrop occurred on three principal occasions, during the Upper Cretaceous, the Upper Miocene (mainly Upper Tortonian) and the Pliocene (see Fig. 2). The stratigraphic relationship of these volcanic horizons is illustrated in an E–W section (Fig. 3). Earlier Mesozoic Volcanic activity is known from sub-surface data.

The most extensive outcrops, and perhaps the best studied, are the Cretaceous outcrops of lavas and dykes exposed at Capo Passero. The flows lie below a limestone sequence dated paleontologically as Maastrichtian (Colacicchi, 1963). Channel et al. (1980), on the basis of the absence of polarity changes, regard them as Campanian. The best radiometric age according to Barberi et al. (1974) is 71 m.y., with results scattered between 71 and 81 m.y., the authors suspecting the presence of excess argon.

The Cretaceous volcanics are, however, considerably more extensive for, in

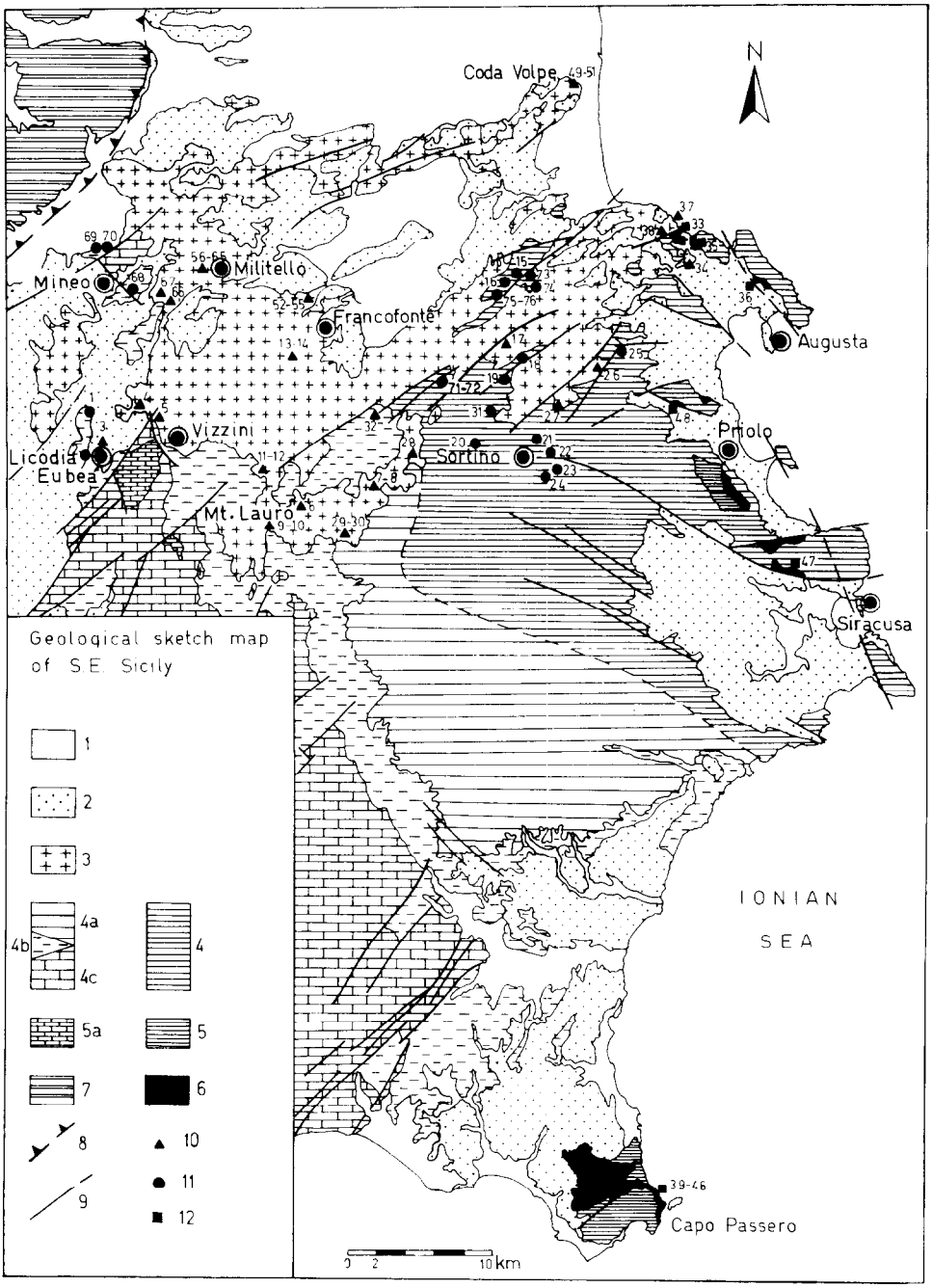


Fig. 2. Simplified geological sketch-map of southeastern Sicily showing the distribution of the sites sampled for paleomagnetic study. *1* = Recent alluvium and beach deposits, *2* = Lower Pliocene and Pleistocene sediments including in the Capo Passero area Messinian clays and evaporites, *3* = Pliocene mafic submarine and subaerial volcanites, *4* = Upper Miocene Sortino Group (full explanation in the text), passing westwards into: Upper Miocene biocalcarenites (Palazzolo Fm.— *4a*), Middle Miocene–top Tortonian marls (Tellaro Fm.— *4b*), Oligocene–Lower Miocene calcarenites and marly limestones (Ragusa Fm.— *4c*), *5* = Upper Cretaceous and early Tertiary reef limestones, *5a* — Upper Cretaceous and early Tertiary cherty marly limestones, *6* = Upper Cretaceous mafic submarine lavas and dykes, *7* = allochthonous Gela nappe, *8* = front of the Gela nappe, *9* = main faults, *10* = sites in Pliocene volcanites, *11* = sites in Upper Miocene volcanites, *12* = sites in Upper Cretaceous volcanites.

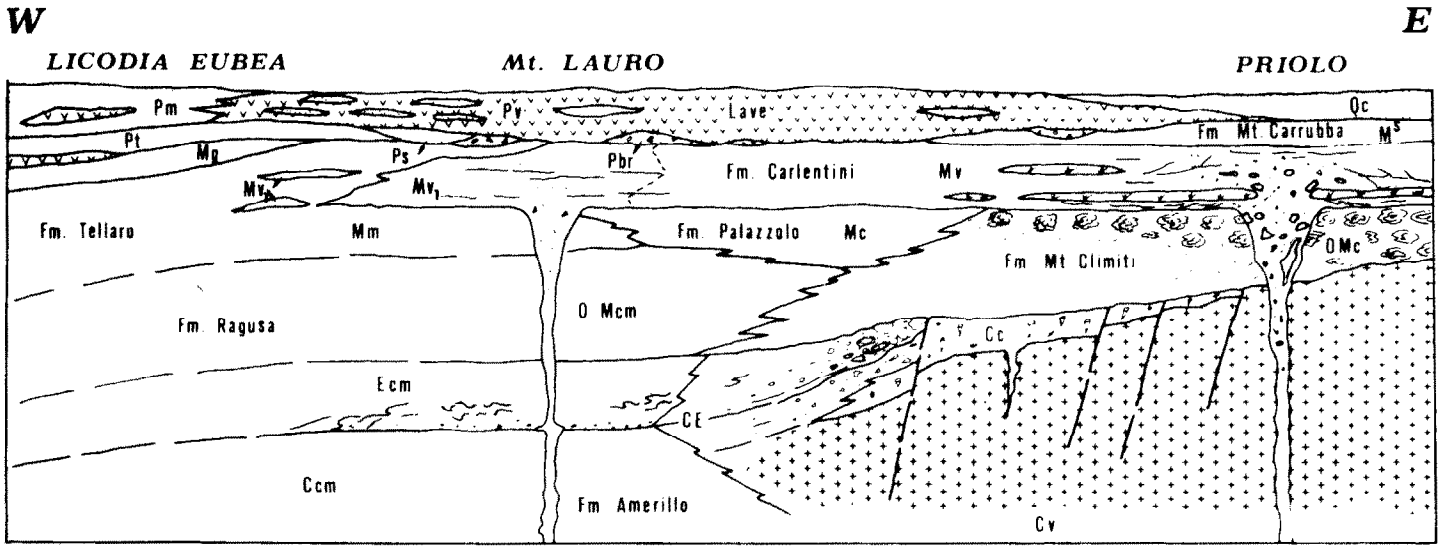


Fig. 3. An E–W schematic cross-section from the Ionian Hyblean coastal areas (Priolo zone) across the Mt. Lauro Plateau and extending into western Hyblea (Licodia Eubea zone) to illustrate the stratigraphic relationship of the volcanic horizons to the enclosing sediments (after Carbone et al., 1982).

The oldest outcropping volcanic horizon is represented by Upper Cretaceous volcanics (*Cv*) overlain by Maastrichtian rudistid reefs (*Cc*), as in the Capo Passero area, or by Oligo-Miocene algal-bryozoan calcarenites of the Mt. Climiti Formation (OMc).

The intermediate level is represented by Upper Miocene volcanics associated with Tortonian bioherms of the Carlentini Formation (*Mv*) or with Tortonian marls of the Tellaro Formation (Mv_1).

The younger level is represented by Pliocene volcanics (*Pv*) overlain, eastwards, by Quaternary sediments (*Qc*). Towards west the volcanics overlie Lower Pliocene breccias (*Pbr*), sands (*Ps*) or "Trubi" (*Pt*). They are also found intercalated with Middle Pliocene sediments (*Pm*). The younger flows of this volcanic sequence reach the Upper Pliocene as results from K-Ar dating (Barberi et al., 1974; this paper) indicate.

The three Mt. Climiti, Carlentini and Mt. Carrubba Formations here shown, have been grouped into the Sortino Group in the geological map of Fig. 2.

addition to the two sites noted by Barberi et al. (1974) near Augusta, several other outcrops are known which extend the area of volcanic outcrops from the vicinity of Siracusa to north of Augusta as well as at Cape Passero.

In the eastern part of the Hyblean Plateau, the Miocene volcanic rocks occur within a shallow water sequence, whereas to the west they are interbedded near the top of a marly sequence (Tellaro Fm.) overlain by Messinian evaporites. Detailed stratigraphic discussion is to be found in Grasso and Lentini (1982) and Grasso et al. (1979, 1982), which are the sources for most of the information presented here.

These Miocene volcanic rocks on the eastern part of the plateau have been assigned to the Carlentini Formation, the middle member of the Sortino Group. The Carlentini Formation lies over algal, bryozoan calcarenites of the Monti Climiti Formation and is in turn overlain by the Monte Carrubba Formation. The surface distribution of the Sortino Group is shown on the geological map (Fig. 2).

In the type section of the Carlentini Formation there is a 45 m section of predominantly volcani-clastic sediments in which a number of flows have been found. The volcanics are divided into three horizons by two interbedded reefs. These biohermal levels have been dated as Upper Tortonian by Grasso et al. (1979, 1982) by means of the coral assemblage and the presence of *Borelis melo*. The Tortonian–Messinian boundary lies within the lower part of Monte Carrubba Formation, for the basal beds contain *Pecten vigolenensis*, a form indicative of the

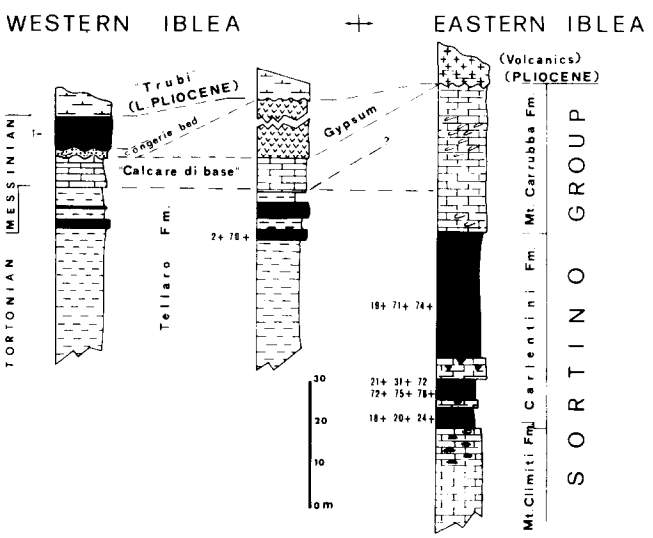


Fig. 4. Stratigraphic correlation of the Carlentini Formation volcanic horizons with the volcanic levels at the top of the Tellaro Formation. The horizons sampled, indicated by site numbers and polarity, correspond to entries in Table I; areal sample distribution is shown in Fig. 1. Sites 16 and 25 cannot be precisely located within the Carlentini Formation and are therefore not shown in the figure. Upper Miocene volcanic levels solid black. Normal magnetization is indicated by " + ", negative by " − ",

Tortonian. They pass from a typical marine to a restricted fauna, a fauna regarded as typical of the Messinian. The age of the Monti Climiti Formation is less well known as the fauna are largely non-diagnostic. Since there is a continuity of sedimentation it must lie within the Tortonian although precise limits cannot be drawn. The Miocene volcanics on the western part of the Hyblean Plateau occur as intercalations at the top of the Tellaro Formation. The formation consists predominantly of clays and marls. In the Vallone Mangalavite, near Licodia Eubea it is capped by a relatively thin limestone, commonly called the "Calcare di base" above which follow Messinian gypsum beds (Fig. 4). According to micropaleontological data (Di Grande and Romeo, 1980), the base of the Messinian occurs within the upper part of the Tellaro Formation, below the level of the volcanic horizons (site 2), a point to which reference will be made in the discussion.

A further volcanic horizon, at Scorciavitelli near Licodia Eubea (Fig. 2, site 1) overlies Upper Messinian sands containing the freshwater *Congerie* fauna. It is in turn overlain by beds of the marine "Trubi", a lower Pliocene formation (Fig. 4).

Although volcanic products occur at various levels in the Pliocene, only hyaloclastites have been found intercalated within Lower–Middle Pliocene sediments. Only the group containing the lowest flows has adequate stratigraphic control. These flows, grouped as the Monte Lauro Plateau group (see Table I) rest upon sands at the boundary between the Lower and Middle Pliocene at some localities near Vizzini and Bucheri. The lowest flows are submarine pillow lavas and represent the infilling of shallow marine basins (e.g. sites 5, 6, 9, 11 and 30) while the succeeding flows are subaerial. North of Sortino the lowest flow (site 27), which rests upon Messinian beds, is already subaerial.

The stratigraphic location of the flow sequences, and of many individual flows is hard to determine, for they are either unconformably overlain by Pleistocene, or the information from the underlying beds is either lacking or indeterminate. Unfortunately there has been relatively little radiometric work and much of that on dikes (see Barberi et al., 1974). This uncertainty has been removed for one flow (site 34) by a K-Ar age reported here, although the errors proved to be relatively large (see Table I).

PETROLOGY

The region of the Iblean Plateau has been the site of sporadic volcanic activity since Mesozoic times. The oldest volcanic rocks found at the surface are Upper Cretaceous in age, older volcanic rocks are known only through boreholes (Cristofolini 1966, 1967b; Patacca et al., 1979). The first recognition of Miocene and Pliocene rocks in the Hyblean Plateau is attributed by Alemagna (1920) to Gemmellaro (1827 and 1833). The principal outcrops are in the northern part of the plateau (Romano and Villari, 1978) where they crop out over an area of 350–400 km^2. They

TABLE I

Summary of the statistical parameters of the paleomagnetic data from the Hyblean volcanic rocks

Sampling *a* denotes the number of samples used in the calculation of the mean data, *b* indicates the number of samples collected. The α_{95} is the semi-angle of the cone of confidence with *K* a measure of dispersion. *R* is the length of the mean vector and under demagnetization is recorded the alternating field or temperature at which the optimum results were obtained.

Localities	Decl. $D(°)$	Incl. $I(°)$	Sampling a/b	α_{95}	K	R	Demag-netization	Notes
	(A) *Summary of paleomagnetic data from Hyblean Pliocene volcanic rocks*							
(1) Francofonte-Militello Lava Group								
Torrente Ossena Flow Sequence								See Honnorez (1962)
14°52′24″E, 37°15′24″N								
Flow 4, site 55	173.3	−63.1	6/6	7.7	76	5.934	300°C	
Flow 3, site 54	164.7	−63.8	6/6	3.4	386	5.987	450°C	
Flow 2, site 53	168.5	−71.8	6/6	9.6	49	5.898	350°C	
Flow 1, site 52	166.1	−67.9	7/7	13.2	22	6.725	400°C	
Militello Flow Sequence								See Honnorez (1962)
14°46′44″E, 37°16′17″N								
Flow 9, site 65A	168.7	−60.2	7/7	5.7	114	6.947	350°C	
Flow 8, site 65	169.1	−75.8	7/8	7.1	74	6.919	400°C	
Flow 7, site 64	164.2	−61.4	5/5	3.0	633	4.994	400°C	
Flow 6, site 63	163.3	−66.5	6/6	3.8	310	5.984	350°C	
Flow 5, site 62	182.5	−64.8	6/7	4.9	186	5.973	400°C	
Flow 4, site 61	175.7	−65.6	6/6	3.8	313	5.984	400°C	
Flow 3, site 60	181.5	−67.9	7/7	4.2	210	6.971	400°C	
Flow 2, site 58	190.9	−58.2	7/7	3.4	311	6.981	400°C	
Flow 1, site 56	187.6	−66.8	6/6	8.1	69	5.928	350°C	
Francofonte Flow Sequence								Site Ib1 of Barberi et al.
14°50′55″E, 37°09′36″N								(1974) K-Ar age: 2.20
Upper flow site 14	182	−55	6/6	6.6	10	5.953	400°C	(±0.60) m.y. (not certain
Lower flow site 13	183.3	−62.1	7/7	3.8	255	6.976	350°C	whether upper or lower flow date)

Monte Lauro Plateau Lava Group								
Vizzini St. pillow lava site 5								
14°43′36″E, 37°09′36″N	156.6	−57.2	5/5	5.1	24	4.983	20 mT	
Buccheri pillow lava site 6								
14°51′18″E, 37°07′23″N	355.7	+46.4	6/6	6.2	117	5.958	20 mT	
Bivio Ferla Flow Sequence								
14°54′40″E, 37°08′06″N								
Upper flow site 7	177.5	−62.9	6/7	7.8	75	5.933	30 mT	
Lower flow site 8	175.8	−58.3	6/7	4.2	214	6.972	65 mT	
Valle Guffari Sequence								
14°49′48″E, 37°06′35″N								
Flow site 10	180.7	−38.3	8/8	4.5	127	7.953	40 mT	
Pillow lava site 9	342.8	+30.3	7/8	7.7	63	6.904	40 mT	
Poggio Morbano Sequence								
14°49′24″E, 37°06′29″N								
Flow site 12	185.7	−51.5	6/6	10.8	40	5.874	65 mT	
Pillow lava site 11	283.9	+53.9	6/7	15.1	21	5.757	20 mT	
Monte S. Germano Sequence								
14°53′28″E, 37°06′07″N								
Upper flow site 29	71.3	+59,8	7/7	10.2	36	6.832	350°C	
Lower spheroidal flow site 30	344.0	+35.3	10/11	7.1	47	9.811	30 mT	
Mt. S. Venere, southern side								
14°56′13″E, 37°09′00″N								
Flow site 28	358.6	+46.5	5/6	6.0	80	4.975	30 mT	
Francofonte–Buccheri road								
14°54′48″E, 37°10′54″N								
Flow site 32	334.9	+59.2	10/11	7.5	42	9.787	30 mT	
(3) C. da Coda Volpe Lava Group								See Honnorez (1962)
15°04′16″E, 37°23′00″N								
Flow 3 site 49	354.5	+46.4	6/6	8.0	71	5.929	350°C	
Flow 2 site 50	12.9	+50.9	6/6	13.0	28	5.819	400°C	
Flow 1 site 51	232.8	+64.6	7/7	12.2	17	6.653	350°C	

TABLE I (continued)

Localities	Decl. $D(°)$	Incl. $I(°)$	Sampling a/b	α_{95}	K	R	Demag-netization	Notes
Individual flows								
Licodia Eubea, flow site								
14°41′56″E, 37°09′04″N	274.9	−11.8	6/6	13.2	26	5.814	30 mT	
Mt Pancali, flow site 17								
15°01′08″E, 37°13′48″N	165	+68.2	11/11	10.9	18	10.456	30 mT	
C. da Malfitano, flow site 26								
15°05′34″E, 37°12′44″N	160.0	−60.0	6/6	4.8	192	5.974	30 mT	
Sortino East, flow site 27								
15°03′24″E, 37°11′10″N	168.8	−70.7	6/6	3.4	384	5.987	40 mT	
Castelluccio South, flow site 34								
15°09′20″E, 37°16′58″N	21.4	+21.5	5/6	8.2	88	4.955	25 mT	K-Ar age: 2.87 ± 0.57m.y 3.80 ± 0.76 m.y.
Baia del Silenzio, flow site 37								
15°08′52″E, 37°18′00″N	352.1	+68.6	7/7	4.1	216	6.972	400°C	
Castelluccio, flow site 38								
14°58′14″E, 37°17′27″N	35.9	+58.4	6/6	3.3	425	5.988	400°C	
Dikes								
Vizzini St., site 4								
14°43′40″E, 37°10′44″N	unstable							
C. da Nicchiara, site 66	unstable							
14°44′40″E, 37°14′58″N								
C. da Annunziata, site 67								
14°43′57″E, 37°14′50″N	194.6	−54.7	6/6	8.1	69	5.928	300°C	Site Ib 18 of Barberi et al. (1974) K-Ar age: 2.85 (+0.14) m.y.

(B) *Summary of paleomagnetic data from Hyblean Upper Miocene volcanic rocks*

(1) Carlentini Fm. flows							
Upper horizon							
Mt. Carrubba site 19							
15°01′08″E, 37°11′10″N	347.2	+46.6	8/8	9.9	32	7.782	15 mT
Carlentini site 74							
15°02′26″E, 37°15′48″N	356.7	+42.8	7/7	6.8	78.5	6.924	400°C
Sorgenti Paradiso site 71							
14°58′08″E, 37°12′08″N	2.8	+41.1	7/7	5.5	122	6.951	300°C
Middle horizon							
Sortino site 21							
15°02′16″E, 37°09′54″N	0.6	+12.5	5/6	7.9	95	4.958	25 mT
Valle Pozzo site 31							
15°00′28″E, 37°11′00″N	358.0	+59.2	7/7	6.2	95	6.937	15 mT
Sorgenti Paradiso site 72							
14°58′08″E, 37°12′08″N	unstable						
Carlentini S.E. site 73							
15°02′20″E, 37°15′58″N	338.4	+68.0	7/7	4.7	104	6.963	500°C
Carlentini S. site 75							
15°01′12″E, 37°15′51″N	14.5	+68	7/7	11.3	29,5	6.797	200°C
Carlentini SW site 76							
15°00′20″E, 37°13′48″N	359.0	+49.8	7/7	8.2	55.4	6.892	300°C
Lower Horizon							
Anapo River site 24							
15°02′32″E, 37°08′24″N	350.1	+35.9	8/8	14.6	15.5	7.541	200°C
Mt. Cugni site 20							
14°59′40″E, 37°09′44″N	353.5	+43.7	6/6	3.4	384	5.987	15 mT
Mt. Carrubba site 18							
15°01′44″E, 37°12′48″N	351.3	+70.8	6/6	6.1	119	5.958	10 mT
(2) Sites of uncertain position							
Carlentini E site 16							
15°01′32″E, 37°16′00″N	195.0	−18.1	5/6	11.0	49	4.918	30 mT
C. da Tenutella site 25							
15°06′24″E, 37°13′27″N	205.9	−36.8	6/6	4.9	192	5.974	450°C

TABLE I (continued)

Localities	Decl. $D(°)$	Incl. $I(°)$	Sampling a/b	α_{95}	K	R	Demag-netization	Notes
(3) Upper Fm. Flows								
Vallone Mangalavite site 2								
14°41'13"E, 37°09'24"N	11.0	+50.0	7/7	15.6	16	6.625	30 mT	
Poggio Impiso site 70								
14°42'18"E, 37°16'48"N	38.9	+66	6/7	7.9	84	5.931	20 mT	
Messinian flows								
C. da Scorciavitelli site 1								
14°38'00"E, 37°10'41"N	146.7	−18.5	5/6	15.0	27	4.852	30 mT	
C. da Signorino site 68								
14°43'18"E, 37°15'07"N	unstable							
(5) Dikes within the Carlentini Fm.								
Carlentini E site 15								
15°02'08"E, 37°16'00"N	193.1	−57.4	6/6	4.9	185	5.943	350°C	
Sortino vent site 22								
15°02'40"E, 37°09'31"N	no result							
Sortino vent site 23								
15°03'14"E, 37°08'34"N	359.5	−6.7	5/5	7.0		4.966	50 mT	
(6) Dikes within the Tellaro Fm.								
Poggio Impiso site 69								
14°42'18"E, 37°16'48"N	158.8	−63.1	4/4	18.4	23	3.872	25 mT	

(C) *Summary of paleomagnetic data from Hyblean Upper Cretaceous rocks*

(1) Capo Passero foreshore sequence 15°08′00″E, 36°41′30″N								Age ranging from 70.7 (±2.0) to 84.1 (±2.4) m.y. (Barberi et al., 1974)
Flow site 40	172.6	−34.3	6/6	3.2	452	5.989	300°C	
Flow site 42	171.3	−28.7	7/7	7.5	65	6.908	300°C	
Dyke site 39	163.9	−43.7	6/6	5.7	139	5.964	300°C	
Dyke site 41	178.3	−35.8	8/8	5.1	119	7.914	300°C	
Dyke site 43	171.2	−8.9	7/7	11.3	30	6.798	400°C	
Dyke site 44	186.1	−37.5	7/7	5.7	113	6.947	400°C	
Dyke site 45	183.2	−25.9	5/5	6.6	134	4.970	300°C	
Dyke site 46	176.6	−37.4	6/6	8.9	57	5.913	300°C	
(2) Siracusa–Augusta outcrops								
C. da Tremilia, dyke site 47 15°14′20″E, 37°04′51″N	162.6	−49,3	7/7	8.5	51	6.883	300°C	
S. Cusumano, dyke site 48 15°09′36″E, 37°11′14″N	181.7	−13.0	5.6	6.6	134	4.970	22.5 mT	77.3 (±1.6) m.y. K-Ar age 77.1 (±1.5) K-Ar age reported by Barberi et al. (1974) for a nearby outcrop: 76.6 (±2.4) m.y.
Mt. Tauro, dyke site 36 15°12′52″E, 37°15′46″N	174.9	+16.8	6/6	32.2	5	5.054	300°C	K-Ar age 81.1 (±2.5) m.y. 78.5 (±2.5) Barberi et al., 1974)
Punta Bonico, lava site 35 15°10′00″E, 36°16′58″N	175.4	−24.5	4/4	14.8	39	3.924	25 mT	K-Ar age: 65.3 (±1.3) m.y. 66.5 (±1.3)
Baia del Gambero, lava site 33 15°09′08″E, 37°17′20″N	166.0	−36.3	6/6	6.4	111	5.955	30 mT	

TABLE II

Summary of the mean paleomagnetic data from Hyblean volcanic rocks

	Sites	Decl.	Incl.	α_{95}	K	Geomagnetic pole position	*dp*	*dm*
Pliocene (37°14'N, 14°56' E)								
Normal flows	10	358.3	+48.6	11.4	19	205.6°E 83.2°N		
Reversed flows	23	174.7	−62.4	3.3	86	344.7°E 82.3°N		
Mean (this paper)	33	356.9	+59.6	5.0	26	339.3°E 85.9°N	5.7	7.5
Barberi et al. (1974)	10	356.0	+48.0	10.0	24	219°E 81°N	9.1	13.9
Gregor et al. (1975)	19	161.0	−50.0	8.3	22	271°E 73°N	8.8	13.1
Overall mean	(3)	351.0	+52.8	12.6	96.2	260.1°E 81.7°N		
Upper Miocene								
Normal flows	13	358.3	+50.8	9.6	20	209°E 84.2°N		
Reversed flows	5	180.6	−41.2	29.1	8	192.2°E 76.5°N		
Mean	18	359.0	+48.3	9.4	15	200.9°E 82.1°N	8.1	12.3
Upper Cretaceous (36°75'N, 14°71'E)								
Schult (1973)	19	167.0	−22.0	7.1	22.8	223.3°E 62.1°N		
Barberi et al. (1974)	8	165.0	−26.0	7.0	62	229°E 63°N	4.1	7.6
Gregor et al. (1975)	15	162.9	−30.1	9.8	13	236°E 64°N	5.0	9.2
This paper	12	173.6	−31.2	7.4	36	212.3°E 69.3°N	4.6	8.3
Overall mean	52	167.3	−27.6	4.3	22	225.1°E 65.2°N	2.6	4.7
	(4)	167.1	−27.4	6.6	193	225.4°E 64.9°N		

range in age from Upper Miocene to Pleistocene (Di Grande, 1972), and can have a total thickness of several hundred metres.

The widespread development of hyoclastites and pillow lavas points to the great extent of submarine volcanism. The pre- and Pliocene volcanics are predominantly marine, and in fact subearial flows become common only when, in Middle Pliocene time, the volcanic pile rose above sealevel. During both Upper Cretaceous and Upper Miocene the submarine volcanoes supported reefs. However, central type volcanoes are rare although many Miocene diatremes filled with breccia can be recognized and even today their morphological forms are clearly visible. In fact the Upper Miocene cycle is made up of the products of phreatomagnetic eruptions (Carbone and Lentini, 1981) with which a few lavas are associated.

Two principal petrological types have been recognized, the tholeiite and alkali basaltic varieties, as in western Sicily, at an earlier stage, and the later olivine basalt and pigeonitic basalts of Carapezza and Morandi (1966). Although Barberi et al. (1974) suggested there was a time difference between the emplacement of the two

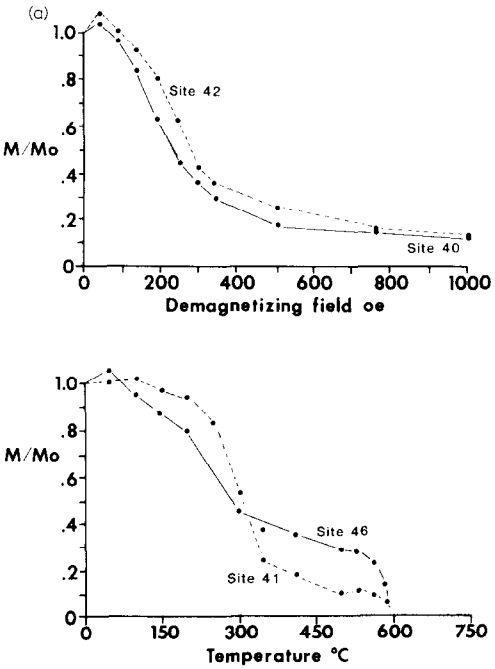
basalt types, this is denied by Romano and Villari (1978) who point to the coexistence of the two lava types and to the lack of any evidence, either stratigraphic or petrologic, to suggest derivation from a common parent magma. There is, however, common agreement that the alkali basalts, volumetrically the more abundant, arose from the partial melting of relatively small amounts of mantle material at greater depths than those at which the tholeiites were generated by the melting of larger volumes of depleted low velocity mantle (Battaglia et al., 1976; Cristofolini et al., 1981; Romano and Villari, 1973).

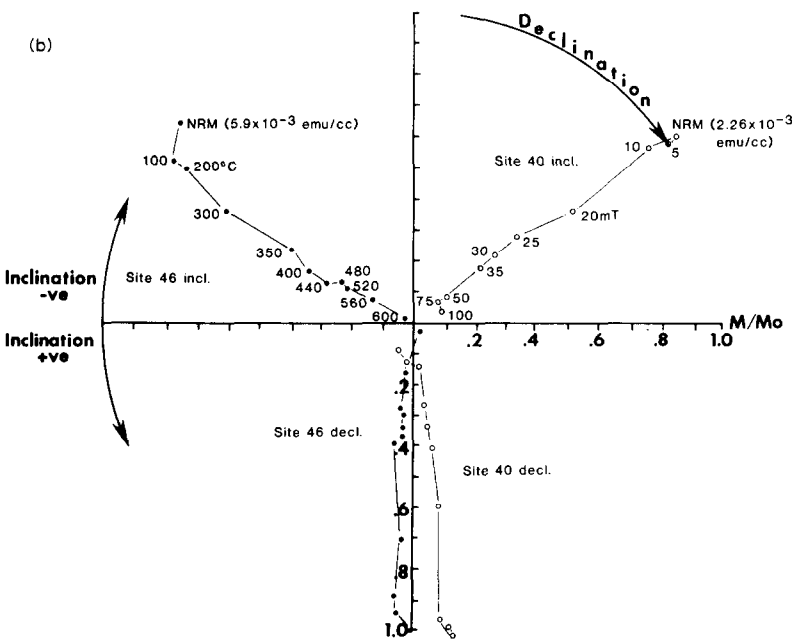
The alkali suite ranges from strongly under-saturated nephelinites to alkali basalts, which in turn can differentiate to hawaiites. The tholeiites are similar to ridge tholeiites but are not regarded by Battaglia et al. (1976) or Cristofolini et al. (1981) as evidence of the existence of a ridge. They suggest that the volcanic activity at the northern edge of the Hyblean Plateau may be linked to the uplift of that region beginning in Upper Miocene time at the end of the most significant phase of plate collision between Africa and Europe.

PALEOMAGNETIC STUDY

All the samples were collected in the field using a gasoline powered field drill and oriented with a Brunton compass. Six cores per volcanic unit was regarded as the optimum; from these one or two specimens could be sliced in the laboratory. The submarine volcanics have usually been substantially altered subsequent to extrusion in a marine environment and the freshest material is usually found in cores of pillows or spheroidally weathering blocks. In some cases, however, the material was so seamed with fractures and joints that cores were not readily obtained. Samples were also obtained from breccias associated with the flows. In the subaerial flows clearly defined vesicular zones with a more compact central zone could often be discerned and samples were usually taken from both areas. From each core one or two samples were cut to permit both thermal and alternating field cleaning. Remanence measurements were made using either the Bologna Schonstedt or the Columbia Digico magnetometers; a comparison of the two instruments showed differences of 2–3° in individual remanence directions. Thermal demagnetization was carried out with a Schonstedt thermal demagnetizer in both laboratories; and for alternating field cleaning a Schonstedt alternating field demagnetizer in Columbia or a locally designed unit in Bologna were used.

The standard routine of detailed demagnetization of pilot samples, by both techniques, to define the optimum stage of demagnetization, followed by batch cleaning at three stages bracketing it, was used. In the succeeding section standard figures, normalized demagnetization curves, stereograms and stable end point curves are given to illustrate the results discussed. These data are summarized in Tables I and II.


(a)
M/Mo
Site 42
Site 40
Demagnetizing field oe
Site 46
Site 41
Temperature °C


(b)
Declination
NRM (5.9x10⁻³ emu/cc)
NRM (2.26x10⁻³ emu/cc)
Site 40 incl.
Site 46 incl.
Inclination -ve
Inclination +ve
M/Mo
Site 46 decl.
Site 40 decl.

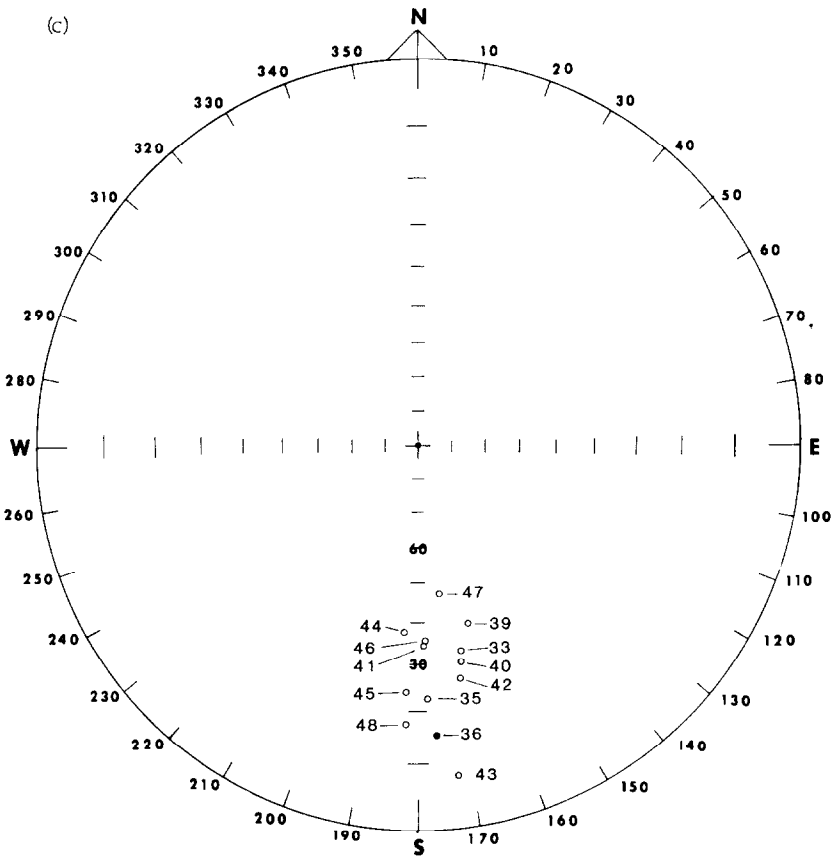
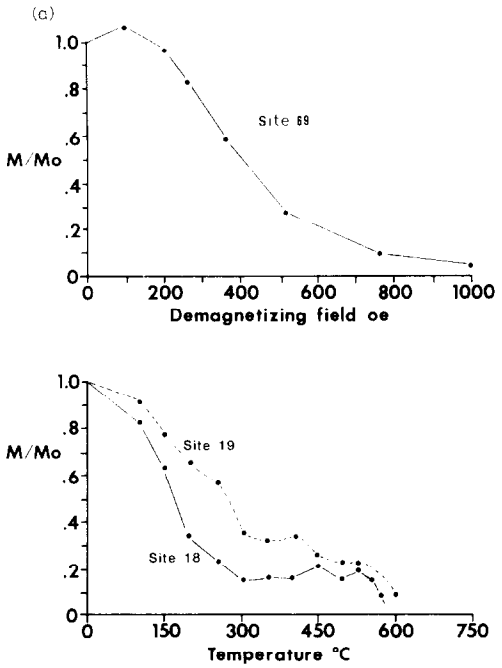


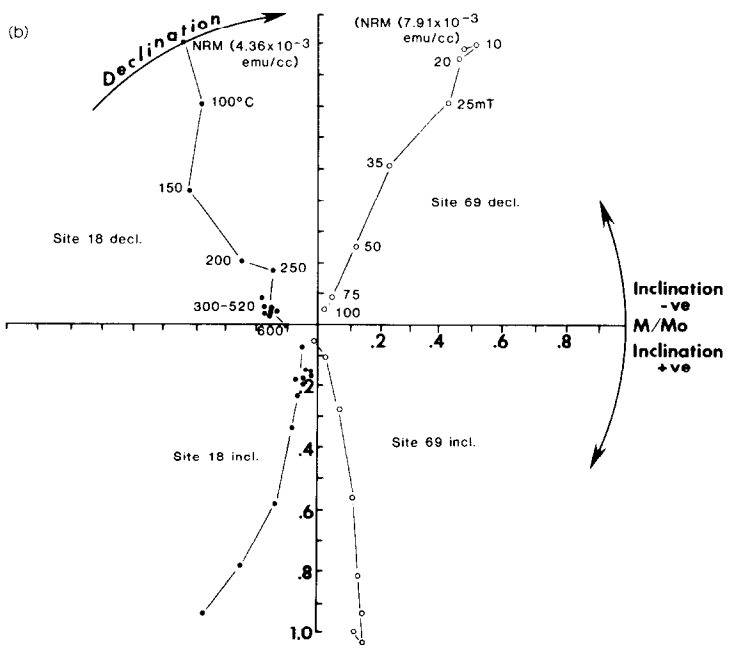
Fig. 5. The results of the examination of the Upper Cretaceous samples. a. Normalized demagnetization curves showing alternating and thermal demagnetization curves of pilot samples. b. Stable end point curves illustrating direction changes during both alternating field and thermal demagnetization of pilot samples. c. Site mean directions of magnetization, stereographic projection (Wulff net).

DISCUSSION

The Cretaceous igneous rocks

From the alternating field cleaning it can be seen that there was a smooth decrease in magnetic intensity (Fig. 5a), whereas the thermal demagnetization indicated on occasion a low blocking temperature of about 300°C which accounted for one half to nine tenths of the natural remanent magnetization. The remaining remanence disappeared at about the Curie temperature of magnetite (see Fig. 5a). The stable end point curves, however do not show significant changes in direction


(a)
M/Mo
Site 69
Demagnetizing field oe
Site 19
Site 18
Temperature °C


(b)
Declination
NRM (4.36x10⁻³ emu/cc)
(NRM (7.91x10⁻³ emu/cc)
100°C
150
200
250
300-520
600
Site 18 decl.
Site 69 decl.
10
20
25mT
35
50
75
100
Inclination -ve
M/Mo
Inclination +ve
Site 18 incl.
Site 69 incl.

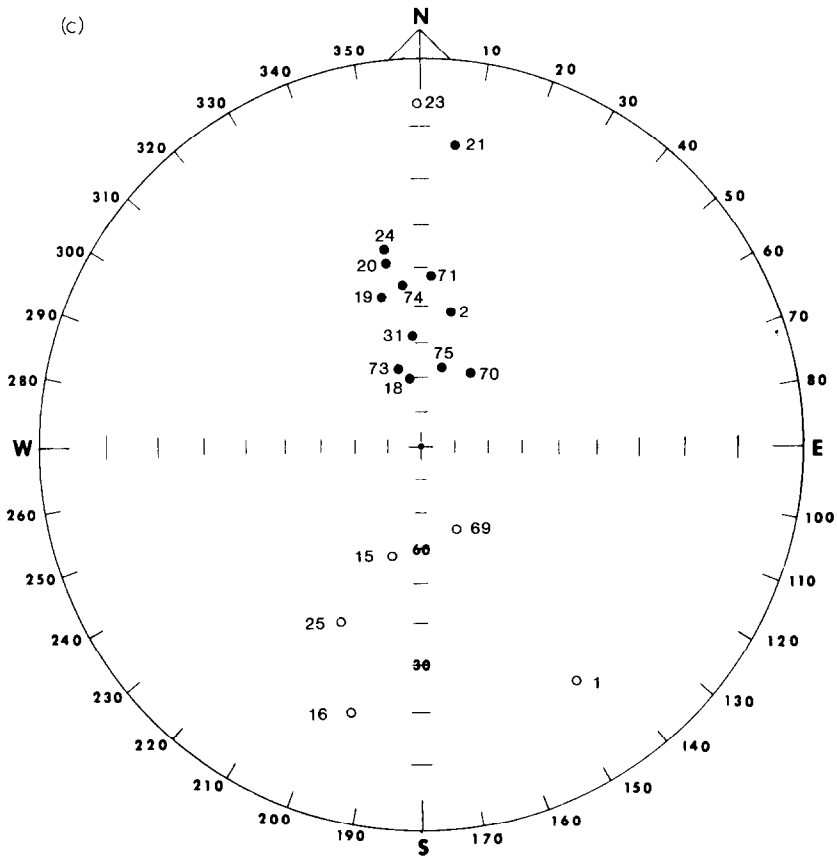
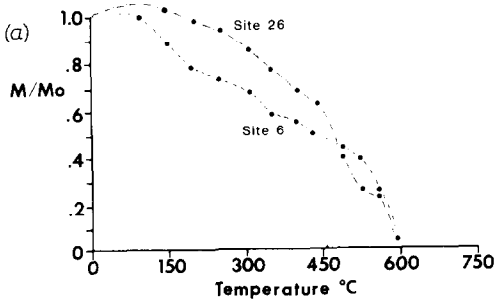
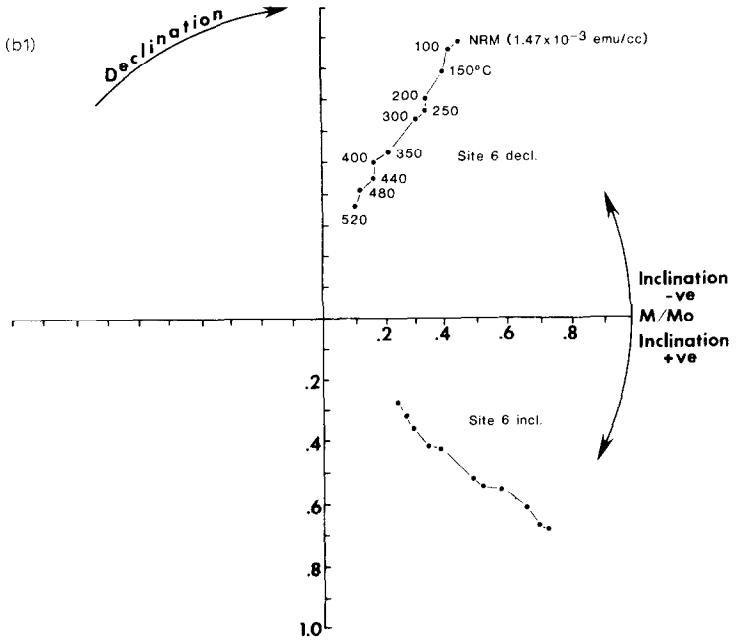


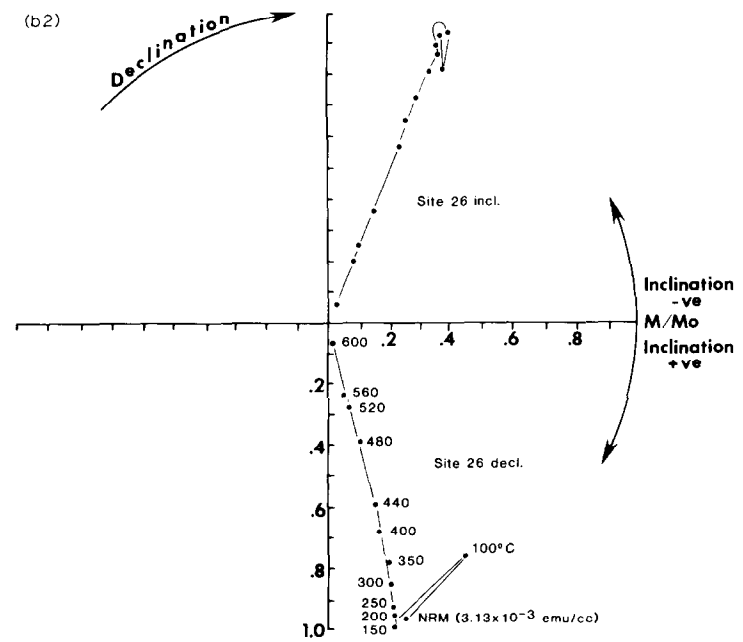
Fig. 6. The results of the examination of the Upper Miocene samples. a. Normalized demagnetization curves showing alternating and thermal demagnetization curves of pilot samples. b. Stable end point curves illustrating direction changes during both alternating field and thermal demagnetization of pilot samples. c. Site mean directions of magnetization, stereographic projection (Wulff net).

during either demagnetization process after the removal of a small viscous component.

It is notable that all the units examined possess a reversed magnetization (Fig. 5c). Given the magnetostratigraphic record of the Cretaceous (Lowrie Alvarez, 1977), this would seem to argue for a burst of igneous activity of relatively short duration. This is not unreasonable if attention is confined to the Capo Passero outcrops, where the dikes and lava flows are intimately associated. Barberi et al. (1974) report the best estimate of the age of the volcanics is 71 m.y. which would place them as Campanian (Channell et al., 1980). The flows lie below a Maastrichtian sequence (Colacicchi, 1963). These same Capo Passero outcrops have been independently sampled by three groups at about the same time, as well as in this study (Table I).


(a)
M/Mo
1.0
.8
.6
.4
.2
0
Site 26
Site 6
0
150
300
450
600
750
Temperature °C


(b1)
Declination
100
NRM (1.47x10⁻³ emu/cc)
150°C
200
250
300
350
400
440
480
520
Site 6 decl.
Inclination
-ve
M/Mo
Inclination
+ve
.2
.4
.6
.8
Site 6 incl.
1.0


(b2)
Declination
Site 26 incl.
Inclination
-ve
M/Mo
Inclination
+ve
600
.2
.4
.6
.8
560
520
480
Site 26 decl.
440
400
350
100°C
300
250
200
150
NRM (3.13x10⁻³ emu/cc)
1.0

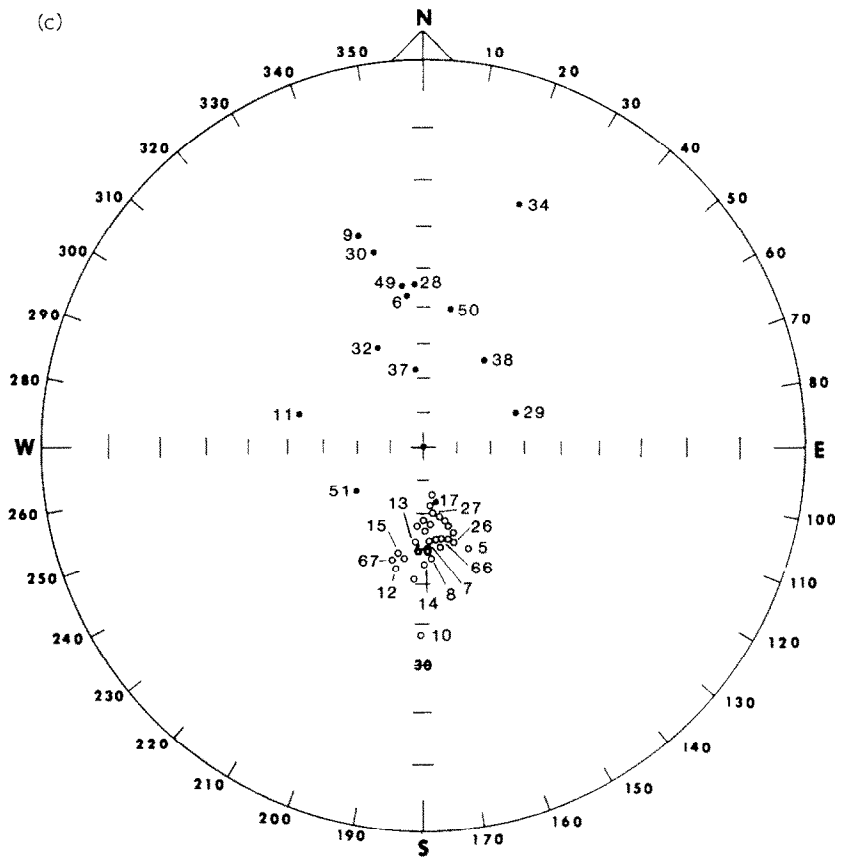


Fig. 7. The results of the examination of the Pliocene samples. a. Normalized thermal demagnetization curves of pilot samples. b. Stable end point curves illustrating direction changes during alternating field demagnetization of pilot samples. c. Site mean direction of magnetization, stereographic projection (Wulff net).

In this study the two northerly sites reported by Barberi et al. (1974) from near Augusta have been augmented by four additional sites which demonstrate that these former seamounts extended at least as far south as Siracusa. Furthermore the radiometric age of the Punta Bonico flow lying close to the Cretaceous–Paleocene boundary raises the possibility of a further phase of volcanic activity.

The mean directions of magnetization from the northern sites lie within the scatter of the Capo Passero sites (Fig. 4c) so that no significant difference can be seen. Thus if there were Pliocene to Recent movements along the Ispica line, the main fault system separating the two areas, are too small to be discriminated paleomagnetically.

The Miocene igneous rocks

The Miocene volcanic rocks are the most varied of the three groups studied. There are a few flows with a dominantly volcaniclastic sequence laid down in a shallow marine environment (Carbone and Lentini, 1981). Pillow lavas and spheroidal weathering flows in general demagnetized to give satisfactory stable end point curves (Fig. 6b). Site dispersion, however, was higher than that found in either Cretaceous or Pliocene rocks.

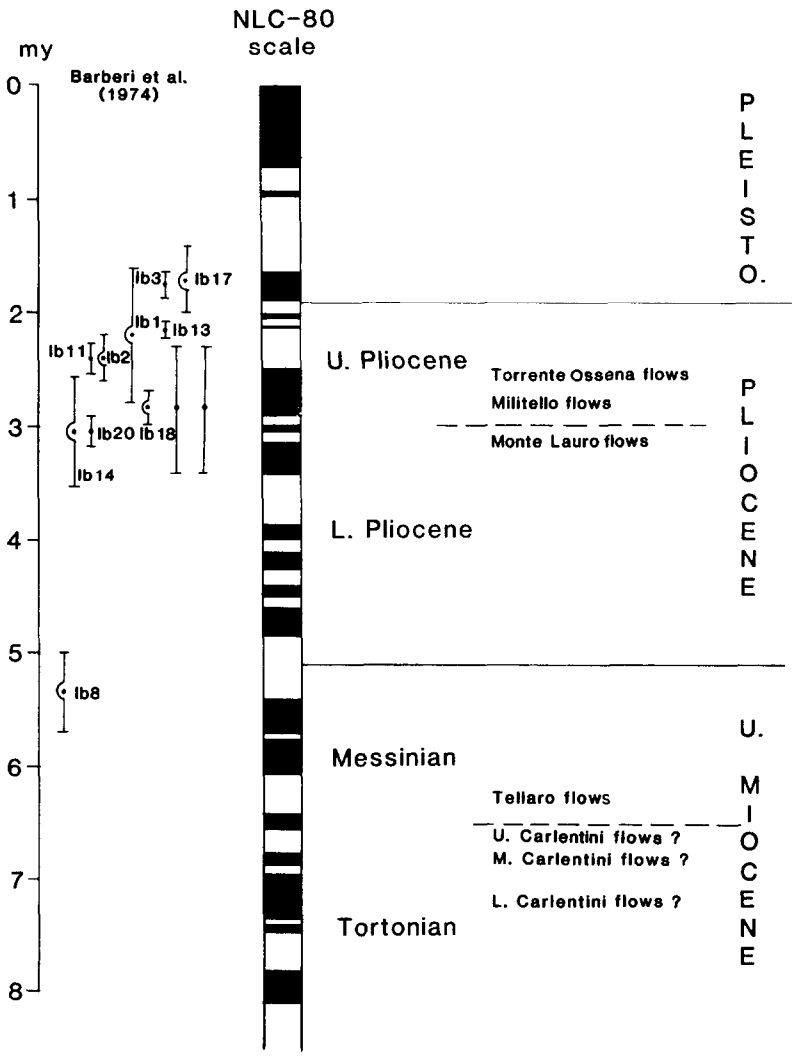


Fig. 8. An attempt to correlate the polarity of the Pliocene and Miocene sites with the magnetostratigraphic record.

The sites reported in this study are all from the Upper Miocene, that is, stratigraphically from rocks of Upper Tortonian and Messinian age. Stratigraphically well defined sites were collected from all three volcanic horizons within the Carlentini Formation; they all possess a normal magnetization. Two sites cannot be located with the same precision, and both of these are reversely magnetized. As the Carlentini Formation is known to be of Upper Tortonian age (Grasso et al., 1979, 1982), we suggest that it equates to magnetic interval 7 (McElhinny 1978) in which the directions of magnetization are predominantly normal. The absence of reversed directions, particularly in the lower igneous horizon suggests that it does not extend below interval 7. The two reversed horizons may represent either of the two reversed events within that interval.

The flows which occur near the top of the Tellario Formation are also normally magnetized. This is inconsistent with a location of the base of the Messinian below that level within the Tellaro Formation, for according to the magnetostratigraphic chart of Ness et al. (1980) a reversed magnetization would be expected. We correlate the flows in the upper part of the Tellaro Formation with those of the Carlentini Formation, thus assigning the top of the sequence to the Upper Tortonian, although this contradicts the paleontological conclusion of Di Grande and Romeo (1980). (See p. 277.)

Results from the four dikes examined are mixed. The one unit (site 69) suggested as related to a particular flow (site 70) proves not to be the case since the two are of opposite polarity. One of the dikes from the Sortino vent (site 22) gave no result and the other (site 23) yielded an anomalous dip. The fourth dike (site 15) reversely magnetized was collected from a location close to one of the two reversed flows found in the Carlentini Formation (site 16) and may well be related to it.

The flow sampled at C. da Scorciavitelli, near Licodia Eubea (Di Grande 1967, 1969a, b), which lies over Upper Messinian sand and below Lower Pliocene Trubi (Fig. 4, site 1) has a reversed magnetization consistent with the location of the Messinian–Pliocene boundary just above the base of the Gilbert polarity epoch.

The Pliocene igneous rocks

Both normal and reversed rocks are found (Fig. 7c). The magnetic behaviour of all samples was relatively simple as Figs. 8a, b illustrate, though dispersion is much greater in the normally magnetized units (see Table I). There are approximately twice as many reversed units as normally magnetized, but this reflects the effect of the long reversed flow sequence of Militello (Honnerez, 1962; Cristofolino, 1967a). There is a number of sites with anomalous magnetization (e.g. site 14).

The lowest flows in the Monte Lauro Plateau Group are pillow lavas (sites 5, 6, 9, 11 and 30), these all prove to have a normal magnetization. The succeeding subaerial flow is reversely magnetized. Given that these flows occur at the Lower–Middle

Pliocene boundary, this is consistent with the pillow lavas being extruded during the Mammoth event.

The long flow sequence at Militello may represent the reversed zone which ends the Matuyama epoch, but the argument is circumstantial and depends upon the tentative correlation of two dated flows of Barberi et al. (1974) with the bottom of the Militello sequence. We tentatively correlate the Torrente Ossena flows with the upper part of the Militello sequence.

Radiometric age dates are relatively few in number and are derived mostly from Barberi et al. (1974). Two of his dated dike sites were re-occupied (Table I) and the polarity re-confirmed. For the majority of the flows, however, there is only the stratigraphic information, which is seldom precise enough to enable the polarity data to be fitted within the magnetostratigraphic time scale.

ACKNOWLEDGEMENTS

The field work upon which this contribution is based was carried out during the sabbatical leave of one of the authors (AEMN). The majority of the measurements were carried out in the paleomagnetic laboratory of the Istituto di Geologia Marina, CNR Bologna. The authors wish to acknowledge gratefully the interest and help given by Dr. M. Manzoni. Field work in Sicily was supported by the Italian CNR through grant 80.01395.89 and by NSF: Int: 77.21521.

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